# Predicting the Next Category and Region of a Visit

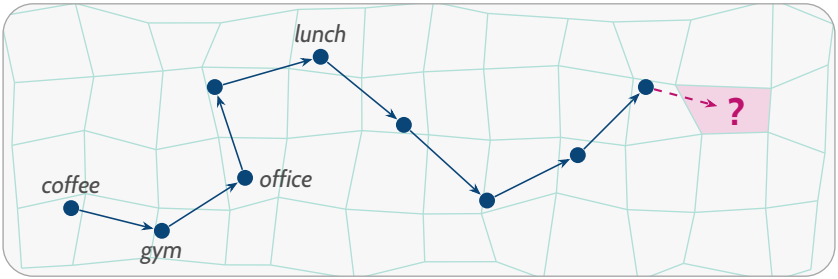
## A Check-in-Level Multi-Task Study on Mobility Data

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# One person's day, and the question




## What kind of place?

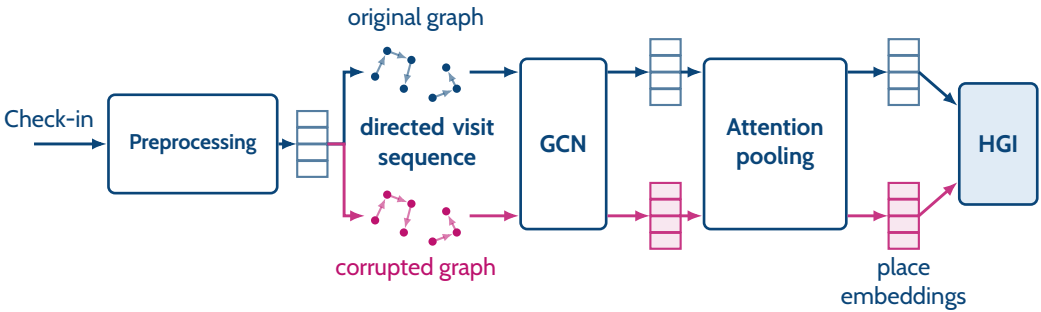
**Next category**: one of 7 types, such as food, shopping, nightlife.

## Which neighborhood?

**Next region**: its census tract, 520 to 8,501 per dataset (Istanbul: *mahalle*).

A tract is a neighborhood, not a radio cell: we claim nothing about cells or handover.

# Check2HGI: a fourth level below the place




- **Each visit carries** its place's category, the hour and day of week, and four elapsed-time values → one **64-d** vector per visit.
- **Built from the past alone**: edges between consecutive visits run forward in time only, and the graph never sees the next category or region.

# Result 1: the representation, at every dataset

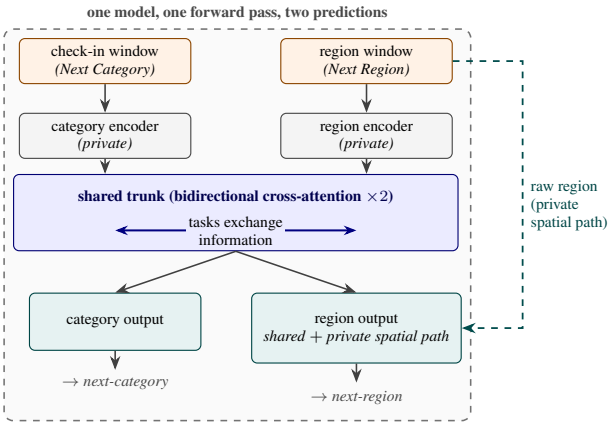
Next category, single-task model, seed 0 — only the input changes

| Dataset | Check-in level | Place level | $\Delta$ |
|---|---|---|---|
| AL | **30.77** $\pm$1.16 | 29.15 $\pm$0.84 | $+1.62$ |
| AZ | **34.51** $\pm$1.15 | 31.93 $\pm$1.08 | $+2.58$ |
| Istanbul | **35.35** $\pm$0.89 | 32.54 $\pm$1.03 | $+2.81$ |
| FL | **37.36** $\pm$0.42 | 37.13 $\pm$0.41 | $+0.23$ |
| CA | **35.62** $\pm$0.47 | 34.74 $\pm$0.41 | $+0.88$ |
| TX | **36.32** $\pm$0.51 | 35.33 $\pm$0.48 | $+0.99$ |

macro-F1; five matched folds, $\pm$ is the fold sd.

**Larger at the three smaller datasets** ($+1.6$ to $+2.8$), under one point at the three largest. A paired test separates the two at five of six; not at Florida.

# Why sharing is not free, and the model




## The risk

Parameters in the middle serve **both jobs at once**, and can settle on a compromise worse for each.

## The answer

A **shared trunk**, where each task reads the other's context; and a **private spatial path** for region that category never touches.

**The private path is the architecture admitting the risk.**

# How it was measured

| | |
|---|---|
| **Regions** | census tracts, **520** (Istanbul) to **8,501** (California) |
| **Split** | by user: the visits of a test user **never appear in training** |
| **Scored** | category: **macro-F1** · region: **Acc@10**, is the true region in the top ten? |
| **Repeated** | **20 trained models** per configuration (4 seeds $\times$ 5 folds) |
| **Decided** | what counts as "not worse": **2 points**, written down before any result was seen |

# One model, two answers: the price

| | **Next region** (Acc@10) | | | **Next category** (macro-F1) | | |
|---|---|---|---|---|---|---|
| | Ded. | Joint | Δ | Ded. | Joint | Δ |
| Alabama | 70.12 | 69.24 | −0.87 | 30.77 | 30.59 | −0.19 |
| Arizona | 59.48 | 59.04 | −0.44 | 34.57 | 34.57 | −0.00 |
| Istanbul | 75.16 | 75.08 | −0.08 | 35.34 | 35.42 | +0.08 |
| Florida | 76.69 | 76.54 | −0.16 | 37.35 | 37.55 | +0.19 |
| California | 63.48 | 64.54 | +1.06 | 35.63 | 35.63 | −0.00 |
| Texas | 64.94 | 66.15 | +1.21 | 36.33 | 36.19 | −0.13 |

Green: outperforms the dedicated model. **Region**: the other four are deficits, not ties, all inside the 2-point margin. **Category**: within a fifth of a point. Δ is the paired mean over the four seeds, so it can differ from the rounded columns in the second decimal.

**The trade is a measured one and not a free substitution.**

# And against published systems

**Next category** (macro-F1)

| | POI-RGNN | Ded. | Joint |
|---|---|---|---|
| AL | 23.80 | 30.77 | 30.59 |
| AZ | 27.64 | 34.57 | 34.57 |
| IST | 30.12 | 35.34 | 35.42 |
| FL | 33.35 | 37.35 | 37.55 |
| CA | 30.71 | 35.63 | 35.63 |
| TX | 32.08 | 36.33 | 36.19 |

**Next region** (Acc@10)

| | HMT-GRN | ReHDM | STAN | Ded. | Joint |
|---|---|---|---|---|---|
| AL | 63.19 | 65.38 | 60.72 | 70.12 | 69.24 |
| AZ | 52.11 | 53.00 | 49.86 | 59.48 | 59.04 |
| IST | 68.06 | 69.33 | 61.86 | 75.16 | 75.08 |
| FL | 72.76 | 64.49 | 72.99 | 76.69 | 76.54 |
| CA | 59.59 | 50.26‡ | 58.61 | 63.48 | 64.54 |
| TX | 61.39 | 48.81‡ | 61.75 | 64.94 | 66.15 |

**Both our models are above every system we compared against**; the joint model by at least **4.11** (category) and **3.55** (region). That margin belongs mostly to the representation, so it is not evidence for sharing.

HMT-GRN: on our windows and folds, read at its best validation epoch, the same rule as our models.
‡ReHDM: its own protocol, single seed at CA/TX.

# What a service would receive

- **Lead time** — median gap from the last visit to the next: **0.4 h** (Florida) to **5.5 h** (Istanbul); 5 to 27% of the targets come more than 3 days later;
- **Unseen users** — every number is on users whose visits never appeared in training;
- **A shortlist** — 10 regions out of **8,501** contain the true next one **64.54%** of the time (California); 10 of 6,553, **66.15%** (Texas).

We built and evaluated no such service: this is motivation, not a measured service result.

# What it costs, and four limits

## THE COST

Parameters, joint vs **the two dedicated models combined**:

AL 4.2 M vs 1.9 M
CA 5.2 M vs 2.8 M

What one model gives is operational: one artifact to train, version and deploy; one pass, two answers.

## FOUR LIMITS

1. Representation trained **once over all users**; a per-fold rebuild moved the dedicated models by at most 0.17 macro-F1 and 0.13 Acc@10;
2. Epoch selection reads the scored fold: **absolute scores are optimistic**; joint vs dedicated is affected far less;
3. Each visit reads **only the visits before it** (slide 2);
4. The joint model is **larger**: its region result may come from size rather than sharing.

# Conclusion

- The evidence **does not attribute the outcome to sharing alone**: on category, the representation moves the score more than the choice between one model and two;
- **One saved model reads both answers**: above the dedicated model on region at Texas and California, within the 2-point margin elsewhere;
- These results do not mean that multi-task learning helps automatically.

What a user will do next **and** where, in a single forward pass instead of two.

# Thank you

Questions?

**Code** (model, representation, baselines, statistical tests)
`github.com/VitorHugoOli/PoiMtlNet/tree/mobiwac`

**Data** (public) · category-annotated Gowalla · Massive-STEPS (Istanbul)

Supported by **FAPEMIG**, **CNPq** and **CAPES**.

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# B1 · Why census tracts

- Predicting over a partition of the map is standard, usually with a **grid cell**; we use **official neighborhood-scale units**;
- Next region is classification over all the dataset's regions:

| | Istanbul | AL | AZ | FL | TX | CA |
|---|---|---|---|---|---|---|
| Regions | 520 | 1,109 | 1,547 | 4,703 | 6,553 | 8,501 |

# B2 · The margin, and the tests

- A written analysis plan, fixed **before any result was read**: **superiority** for category, **non-inferiority** for region;
- Region margin $\delta = 2$ **Acc@10 points**, fixed in advance; two one-sided tests (TOST);
- Paired $t$ on the four per-seed means, 90% interval, **Holm** correction across the six datasets, per task;
- Category has **no registered equivalence margin**: a difference that fails superiority is unresolved, never a match;
- The two region gains are **secondary results**, outside the registered plan.

# B3 · The Markov floor

- A first-order Markov chain over region transitions, no learning: **51.23 to 72.47** Acc@10;
- The joint model exceeds it by $+4.1$ **to** $+10.0$ at every dataset; the narrowest is **Florida,** $+4.07$.

# B4 · Can a visit vector see its target?

- Each visit node draws on the **visits that precede it**: the vector encodes the visit with its own history;
- The graph does **not pass information from a later visit back to an earlier one**, in training or at readout;
- That is what keeps a node from carrying a feature of the target it is used to predict;
- The graph is trained without the next-category or next-region label.

# B5 · One model instead of two

- **4.2 M vs 1.9 M** parameters at Alabama, **5.2 M vs 2.8 M** at California, against **the two dedicated models combined**;
- One pass costs more compute than running the two small models;
- The gain is operational: one artifact to train, version and deploy; one pass returns both answers.

# B6 · Then what is multi-task learning for?

- On category, the representation moves the score **more than the choice between one model and two**;
- Whether the exchange between the tasks adds anything beyond the representation is not separated by the evidence here;
- What one model gives for certain is operational: one artifact, one forward pass.

# B7 · The external systems

- **HMT-GRN** — primary region comparison: our windows and folds, one seed, best epoch; its next-place components removed;
- **STAN** — re-implemented with its own representations, output adapted to regions; its own embeddings and splits, seed 0;
- **ReHDM** — its own published chronological split; single seed at TX and CA;
- **POI-RGNN** — next category, re-implemented, non-overlapping windows.

# B8 · More regions, more gain?

| | Regions | Region $\Delta$ (Acc@10) |
|---|---|---|
| Texas | 6,553 | $+1.21$ |
| California | 8,501 | $+1.06$ |

California has more regions than Texas and a **smaller** gain.

An observation, not a law.

# B9 · Is it just the raw features?

Control: the place vector + the same raw per-visit features (category one-hot, hour, day of week), same single-task model.

| | Control gain | Check-in-level gain |
|---|---|---|
| Alabama | $+1.73$ | $+1.62$ |
| Arizona | $+1.70$ | $+2.58$ |
| Florida | $+1.02$ | $+0.23$ |

macro-F1 over the place-level representation, category axis.

*"Most of the category difference is therefore already available in the raw per-visit features, and this control does not separate what the check-in-level representation adds beyond them on this axis."*

# B10 · The intervals

Joint minus dedicated, 90% interval of the paired difference

| | **Next region** (Acc@10) | | **Next category** (macro-F1) | |
|---|---|---|---|---|
| | $\Delta$ | 90% CI | $\Delta$ | 90% CI |
| Alabama | $-0.87$ | $[-1.00, -0.75]$ | $-0.19$ | $[-0.33, -0.04]$ |
| Arizona | $-0.44$ | $[-0.62, -0.25]$ | $-0.00$ | $[-0.04, +0.03]$ |
| Istanbul | $-0.08$ | $[-0.16, -0.002]$ | $+0.08$ | $[+0.01, +0.15]$ |
| Florida | $-0.16$ | $[-0.19, -0.13]$ | $+0.19$ | $[+0.14, +0.25]$ |
| California | $+1.06$ | $[+1.03, +1.08]$ | $-0.00$ | $[-0.03, +0.02]$ |
| Texas | $+1.21$ | $[+1.13, +1.29]$ | $-0.13$ | $[-0.19, -0.08]$ |

**Green**: outperforms, Holm-corrected — region at Texas ($p = 0.00013$) and California ($p < 10^{-4}$), 20 of 20 folds; category at Florida ($p = 0.011$), 19 of 20.

Every region interval of the other four lies **entirely below zero**: deficits, not ties.